

Shangjie (Henry) Zheng

(816) 437-5658 | shangjiehz@gmail.com | www.shangjiezheng.com

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, C++, C, SQL, MATLAB, JavaScript, TypeScript, HTML, CSS
- **Databases & Data Streaming:** MySQL, PostgreSQL, SQLite, MongoDB, Amazon S3, Redis, Kafka, BullMQ, Hibernate
- **Frameworks & Platform:** Spring Boot, Django, React, Node.JS, Express, AWS, Linux, Zephyr RTOS, CAN, J1939
- **Tools & Libraries:** Docker, Kubernetes, Terraform, Git/GitHub, Postman, Maven, Apache JMeter, Nginx, spaCy, NLTK

PROFESSIONAL EXPERIENCE

Lion Energy LLC. American Fork, UT
Embedded Systems Engineer, Controls Software | C, Zephyr RTOS, MATLAB, Simulink, J1939 Jan. 2026 – Present

- Architected and implemented a modular embedded application layer in C on Zephyr RTOS for Energy Storage Systems (ESS), enabling seamless system-level control of up to 6 power distribution units through deterministic task scheduling and structured fault management.
- Developed and optimized a CAN-based messaging subsystem (J1939) serving as a unified communication service layer across 7+ ESS functional domains, enabling distributed peer-to-peer and broadcast communication, multipacket transport, and OTA update capabilities.
- Spearheaded the implementation of a SIL/HIL validation framework by integrating production firmware modules into Simulink test harnesses, enabling closed-loop simulation, hardware abstraction and repeatable regression testing of real-time control algorithms.

Southern Methodist University Dallas, TX (Remote)
Research Collaborator | Python, GPT, Gemini, Keras, spaCy Aug. 2023 – Present

- Developed and refined a machine learning toolset aiming to translate tax domain-specific knowledge into code-like representations.
- Fine-tuning GPT and Gemini LLMs with pre-annotated tax documents by classifying contexts into (non-)environmental variables, leveraging Keras, spaCy libraries to develop a specialized classifier that accurately interprets domain-specific language.
- Implemented programs to automate the data collection and sanitization, populate datasets, and validate raw internet data sources.

Lion Energy LLC. American Fork, UT
Control Systems Engineer | C, MATLAB, Simulink, J1939, Zephyr RTOS Jul 2024 – Jan. 2026

- Designed and optimized software components for Energy Storage System, focusing on reliability and performance improvements.
- Led the design and implementation of a DSP module within the real-time telemetry pipeline, improving aggregate data accuracy by 35%.
- Built a MATLAB-based test automation framework integrated with Simulink Test API, reducing manual test creation time by 90%.

Lion Energy LLC. American Fork, UT
Control Systems Engineer Intern | C, MATLAB, Simulink, MCU, CAN Jul 2024 – Dec. 2024

- Collaborated in the development and research of embedded software for Battery Management System (BMS).
- Refined SoC estimation through end-of-charge/discharge state Amp-Volt mapping optimization, increasing accuracy by over 30%.
- Redesigned cell balancing algorithm to discharge optimal balancing current from target cells, speeding process from days to hours.

SELECTED PROJECTS

Riivr App, Distributed Backend Social Platform American Fork, UT
Node.js, TypeScript, MongoDB, Redis, BullMQ, AWS (EC2, S3, ElastiCache, CodeDeploy), Terraform Spring, 2026

- Architected a domain-driven backend spanning 9 bounded contexts, 60+ REST endpoints, and 10 persistence models, integrating JWT authentication in a layered pipeline (routes, middleware, controllers, services), enforcing modular boundaries and validations.
- Developed a cache-first Redis-over-MongoDB strategy, reducing latency and database pressure with write-through consistency.
- Leveraged BullMQ-backed asynchronous job queues to offload non-blocking workflows and decouple background processing.
- Provisioned AWS infrastructure via Terraform across 15+ resources (multi-AZ, ALB/HTTPS, EC2, NAT, Bastion, ElastiCache).

Shop-Core, High-Concurrency E-Commerce Backend System American Fork, UT
Spring Boot, Redis, MySQL, MyBatis, Kafka, JMeter, Thymeleaf, Nginx Fall, 2024

- Built a high-concurrency e-commerce backend using Spring Boot, leveraging Redis-over-MySQL pipeline to increase QPS 20x.
- Integrated Kafka message queue to decouple Redis single-thread handling from incoming message rates and mitigate peak loads.
- Conducted comprehensive unit testing with Junit achieving 76% overall code coverage, and performed load tests using JMeter.

EDUCATION

Southern Methodist University (SMU) May 2024
Master of Science in Computer Science GPA: 3.97/4.00

University of Missouri-Kansas City (UMKC) May 2021
Bachelor of Science in Mechanical Engineering GPA: 3.52/4.00